



2.5 Exponential and logarithmic functions

Functions · IB Diploma · Concept guide and original exam-style practice

Understand the concept

An exponential function $f(x) = a \cdot b^x$ ($b > 0$, $b \neq 1$) has a horizontal asymptote $y=0$, is always positive if $a > 0$, and models growth ($b > 1$) or decay ($0 < b < 1$). Its inverse is a logarithmic function $f^{-1}(x) = \log_b(x/a)$, which has domain $x > 0$, a vertical asymptote $x=0$, and is the reflection of the exponential graph in $y=x$. The natural exponential e^x and natural logarithm $\ln x$ are the special case $b=e$.

Worked example: follow the reasoning

Sketch key features of $y = 3^x - 2$ and state its horizontal asymptote and y-intercept.
As $x \rightarrow -\infty$, $3^x \rightarrow 0$, so $y \rightarrow -2$: horizontal asymptote $y = -2$.
Y-intercept ($x=0$): $y = 3^0 - 2 = 1 - 2 = -1$.
Since $3^x > 0$ always, $y > -2$ for all x , so range is $y > -2$.

Quick check · multiple choice

What is the horizontal asymptote of $y = 5e^{-x} + 1$?

- A $y=0$
- B $y=1$
- C $y=5$
- D $x=0$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Consider the function $f(x) = 2e^{(x-1)} - 3$. [7]
- (a) State the equation of the horizontal asymptote of $y=f(x)$.
 - (b) Find the exact coordinates of the y-intercept.
 - (c) Find $f^{-1}(x)$ algebraically and state its domain.

Final answer: _____



Answers and explanation

Question 1 [7]

(a) As $x \rightarrow -\infty$, $e^{(x-1)} \rightarrow 0$, so $y \rightarrow -3$: asymptote $y = -3$.

(b) $f(0) = 2e^{(-1)-3} = 2/e - 3$.

(c) $y = 2e^{(x-1)-3} \rightarrow (y+3)/2 = e^{(x-1)} \rightarrow \ln((y+3)/2) = x-1 \rightarrow x = 1 + \ln((y+3)/2)$. So $f^{-1}(x) = 1 + \ln((x+3)/2)$, domain $x > -3$.

Quick check answer

B. As $x \rightarrow \infty$, $e^{(-x)} \rightarrow 0$, so $y \rightarrow 1$; this is the horizontal asymptote.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.